

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

28
100

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: C&A15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct (192324017)

I S. Aswini Surya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

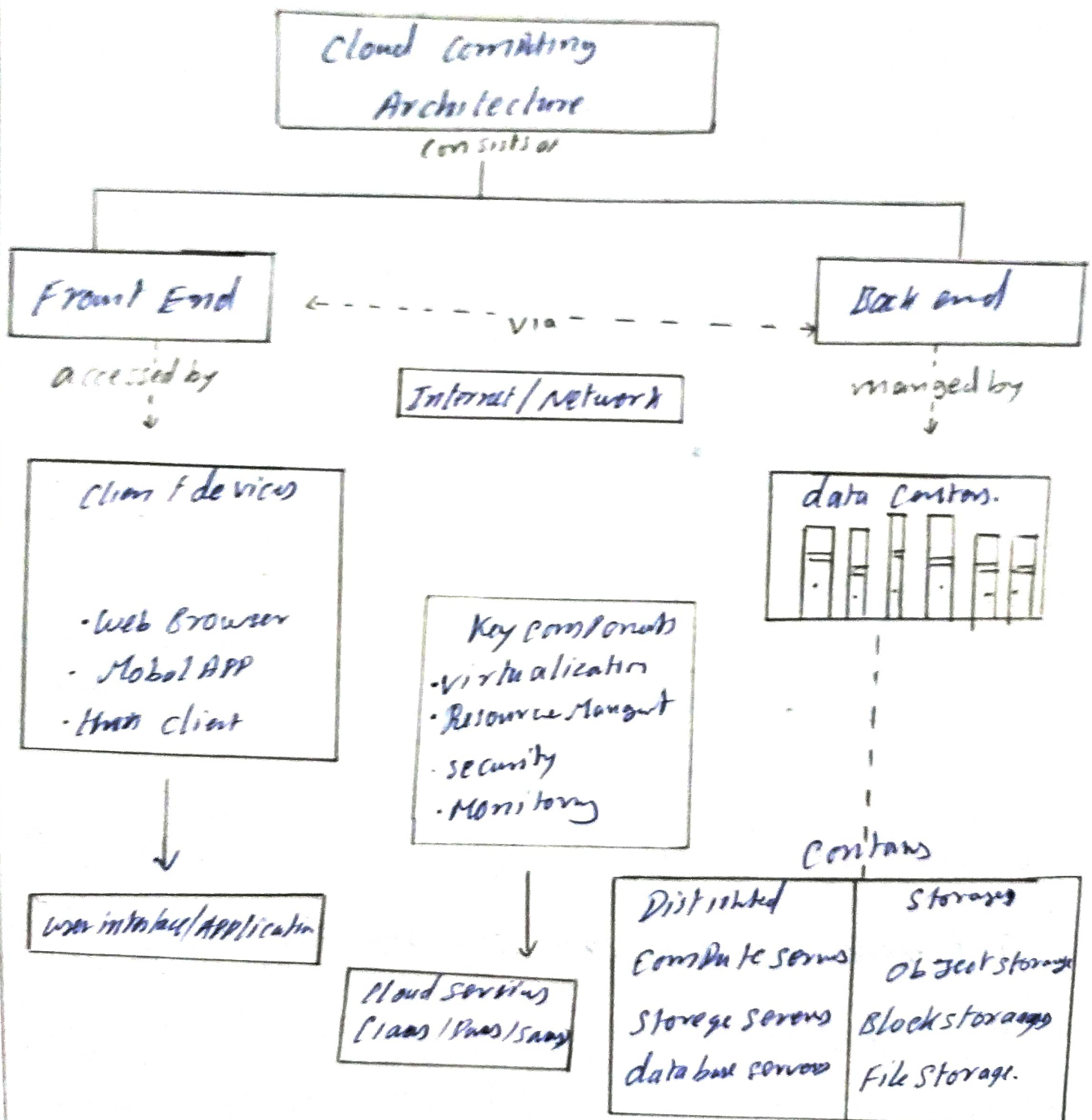
Signature of the student: SAS

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

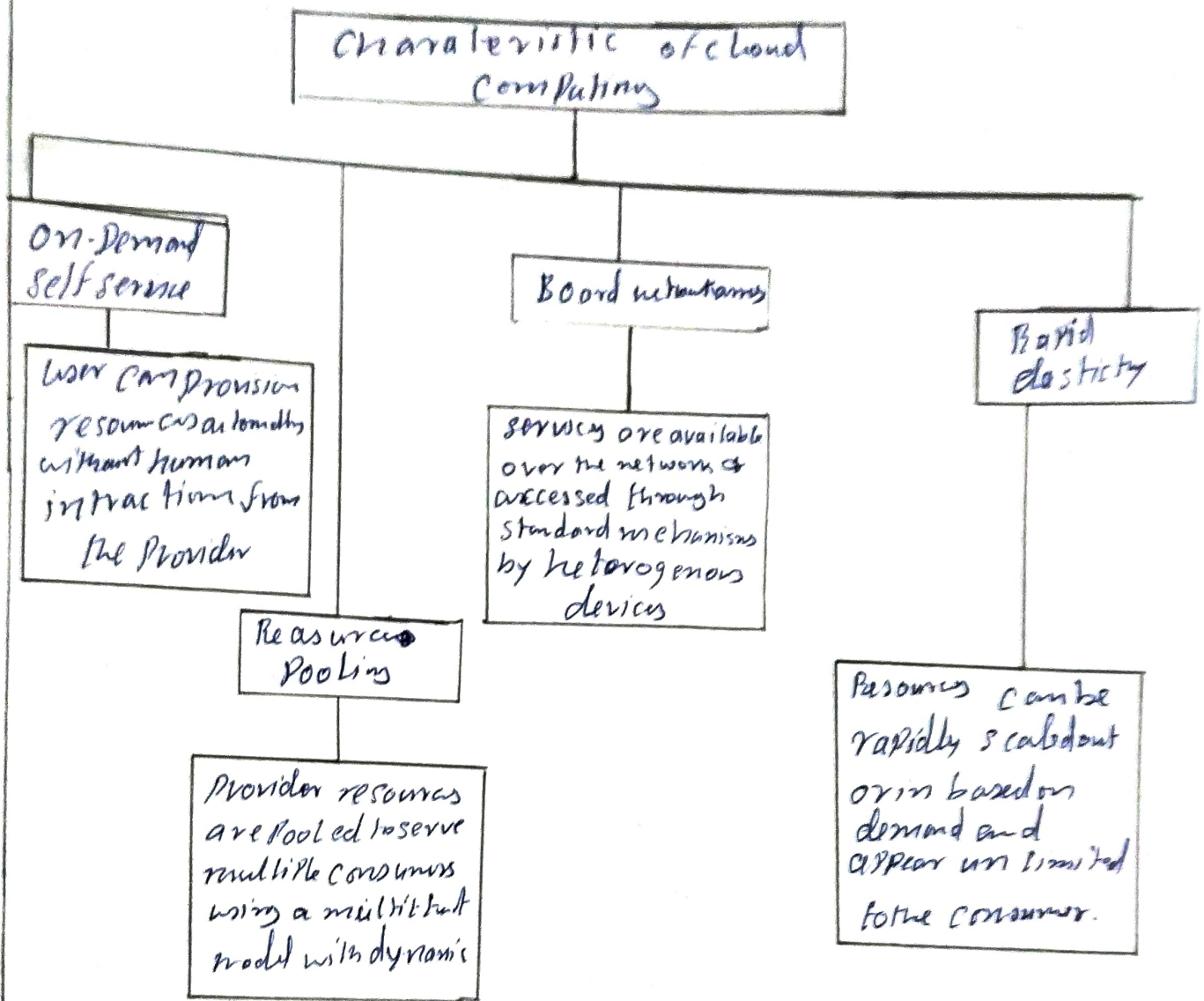
Signature of the Faculty

Total Marks Awarded:

1. Concept map: cloud computing Architecture include front end, Backend client devices, data centers and distributed servers

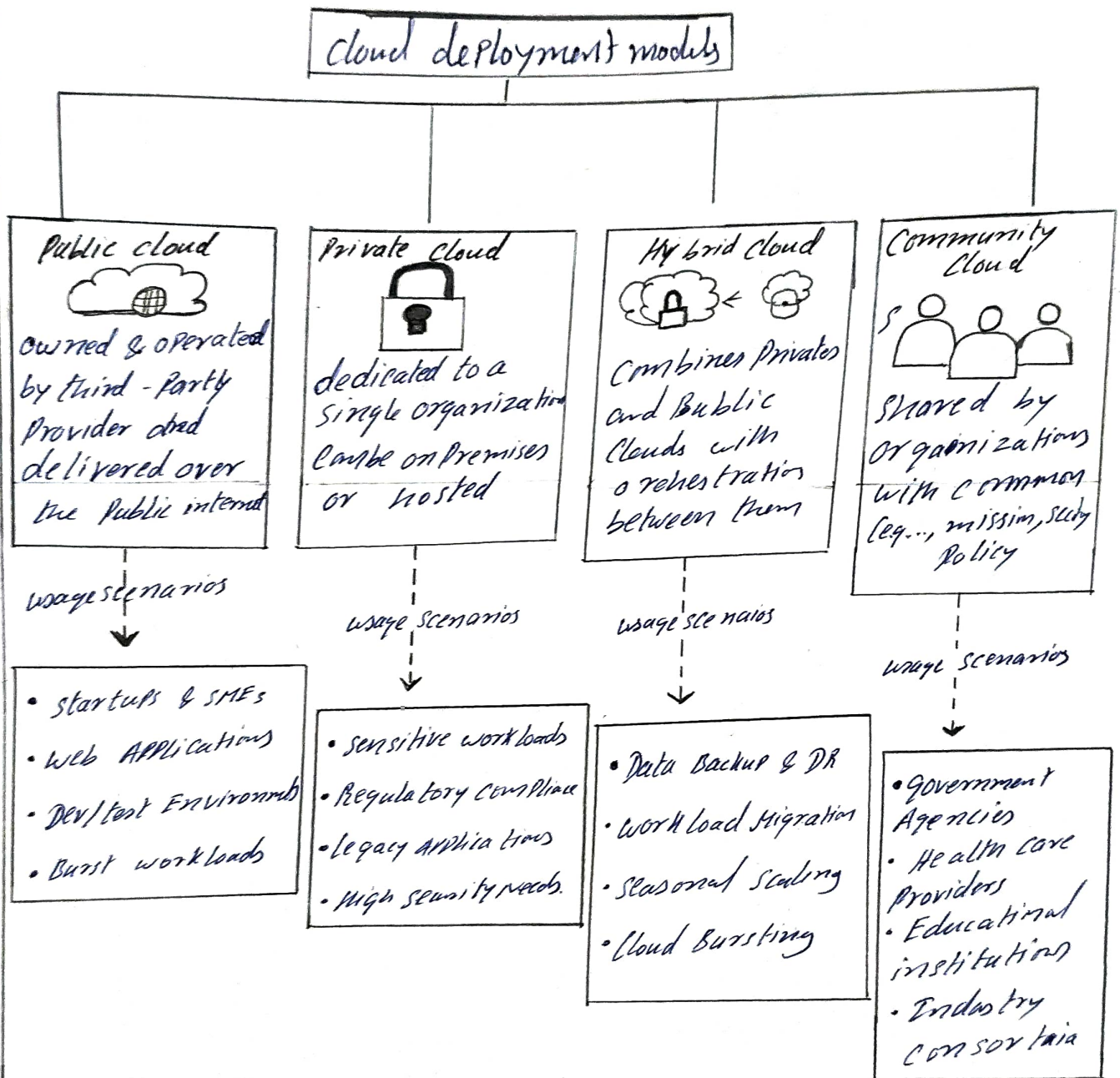


2. Concept Map 2: characteristics of cloud computing
- Connect on demand self service, broad network access, resources pooling rapid elasticity and measured service



Measured Service: Resource usage is metered, monitored and reported for transparency and pay-as-you-go model.

concept map: Deployment Models: Map Public cloud, Private cloud, hybrid cloud and community cloud with usage scenarios.



Common Benefits Across All Models
 Cost Efficiency / Scalability / Flexibility / Improved collaboration

Cloud Computing

Benefits/Advantages

Cost Efficiency



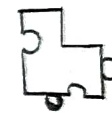
Lower upfront costs, pay-as-you-go pricing reduces maintenance.

Scalability



Easily scale resources up or down based on demand and

Flexibility



Access services anytime any where; supports innovation and agility

Additional Benefits
High Availability (recovery time to market)
Global reach / Disaster Recovery

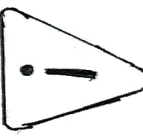
Challenge / Limitations

Security Concerns



Data breaches, privacy issues, shared responsibility risk.

Downtime / Reliability



Outages can impact business operations and SLAs

Vendor Lock-in



Difficulty in migrating data and applications to another provider

Additional challenges
Compliance issues / limited control
Performance variability / hidden costs

Key to Success

Proper Planning / Risk Management / Provider Evaluation / Governance

4. Concept map 1: cloud benefits vs challenges: show advantages (cost efficiency, scalability, flexibility) and limitations